Jordan Blocks and Defective Matrices

Exam notes for repeated eigenvalues, missing eigenvectors, and Jordan form.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Repeated eigenvalues do not guarantee enough eigenvectors. When a matrix is missing eigenvectors, Jordan blocks replace diagonal form.

Diagonalization is the best case:

$$A = S\Lambda S^{-1}.$$

This works only when A has a full basis of independent eigenvectors.

2 Diagonalizable vs Defective

Case	Meaning	Exam signal
Diagonalizable	Enough independent eigenvectors to form S.	Each eigenspace supplies the needed number of vectors.
Defective	Not enough independent eigenvectors.	Some repeated eigenvalue has too small a nullspace.
Jordan form	Replacement for diagonal form.	Ones appear just above the diagonal inside a block.

3 Algebraic and Geometric Multiplicity

For an eigenvalue λ:

$$m_a(\lambda) = \text{multiplicity of } \lambda \text{ in } \det(A - \lambda I),$$
$$m_g(\lambda) = \dim N(A - \lambda I).$$

The exam test is:

$$A \text{ is diagonalizable} \quad \Longleftrightarrow \quad m_g(\lambda) = m_a(\lambda) \text{ for every } \lambda.$$

If $m_g(\lambda) < m_a(\lambda)$, the matrix is defective for that eigenvalue.

4 Jordan Blocks

A Jordan block for eigenvalue λ has λ on the diagonal and ones on the superdiagonal:

$$J_\lambda = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}.$$

The diagonal part gives the eigenvalue. The superdiagonal ones record the missing eigenvector directions.

5 Jordan Form

Instead of diagonalizing, a defective matrix may be written as

$$A = MJM^{-1},$$

where J is block diagonal with Jordan blocks.

A diagonal matrix is the special case where every Jordan block has size 1.

6 Generalized Eigenvectors

If x is an eigenvector, then

$$(A - \lambda I)x = 0.$$

A generalized eigenvector w may satisfy

$$(A - \lambda I)w = x.$$

Together, w and x form a chain. This chain supplies the columns of M for the Jordan block.

7 Exam Method

1. Find eigenvalues from $\det(A - \lambda I) = 0$.
2. For each repeated eigenvalue, compute $N(A - \lambda I)$.
3. Compare $m_g(\lambda)$ with $m_a(\lambda)$.
4. If all match, diagonalize.
5. If one fails, use Jordan blocks for that eigenvalue.

8 Common Mistakes

Mistake	Correct exam habit
Assuming repeated eigenvalues mean defective.	Check the dimension of the eigenspace.
Assuming distinct eigenvalues are required.	Repeated eigenvalues can still be diagonalizable.
Counting algebraic multiplicity as eigenvectors.	Count independent vectors in $N(A - \lambda I)$.
Forgetting the superdiagonal ones in a Jordan block.	A nontrivial block has ones above the diagonal.
Treating $A = MJM^{-1}$ as diagonalization.	Jordan form is weaker than diagonal form.

9 Final Exam Checklist

Checklist.

1. Compute the characteristic polynomial.
2. List algebraic multiplicities.
3. Compute eigenspace dimensions.
4. Compare $m_a(\lambda)$ and $m_g(\lambda)$.
5. Use diagonal form if enough eigenvectors exist.
6. Use Jordan blocks when eigenvectors are missing.
7. Check that the total block sizes add to n.